



7014 - Searching for Population III stars around massive interacting galaxies in the reionization epoch

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Pointing 1				
	1		NIRSpec IFU Spectroscopy	(1) ID_1
Pointing 2				
	2		NIRSpec IFU Spectroscopy	(2) ID_2
Pointing 3				
	3		NIRSpec IFU Spectroscopy	(3) ID_3
Pointing 4				
	4		NIRSpec IFU Spectroscopy	(4) ID_4_5
Pointing 5				
	5		NIRSpec IFU Spectroscopy	(5) ID_6
Pointing 6				
	6		NIRSpec IFU Spectroscopy	(6) ID_7
Pointing 7				
	7		NIRSpec IFU Spectroscopy	(7) ID_8
Pointing 8				
	8		NIRSpec IFU Spectroscopy	(8) ID_9

ABSTRACT

Recently, the efforts to uncover traces of the elusive Population III stars have intensified, with several studies hinting at their detection in early galaxies. Yet, definitive observational evidence of this stellar population is still lacking. We propose a new, physically motivated approach, complementary to existing surveys, to search for Pop III stars by observing with NIRSpec-IFU (at medium resolution) the environment of 9 massive and interacting galaxies between redshift 7 and 9.5, which recent simulations suggest may host rich Pop III-forming environments in their surroundings with high probability. Pop III stars will be clearly identified by the presence of bright HeII 1640 line and Balmer lines, and faint or absent metal lines, supported by robust photoionization model predictions. Additionally, these observations will provide new insights into the star formation and interstellar medium properties of merging galaxies at $z > 7$, improving our understanding of galactic interactions during this early

epoch.

OBSERVING DESCRIPTION

This program involves NIRSpec-IFU observations of 9 galaxies in the redshift range between 7 and 9.4. The objective is to search for Population III stellar clumps around massive galaxies, maximizing the probability of finding them, according to recent simulations. This will be achieved by analysing their rest-frame UV and optical lines, through which we can clearly distinguish between photoionization from metal poor Pop II stars and metal free stars. We thus request to combine observations with the medium resolution gratings G140M and G395M.

The galaxies lie in the EGS field and around the cluster Abell 2744. All of them have exquisite, multiband NIRCам images, and were confirmed spectroscopically through previous NIRSpec Multi-object spectroscopic (MOS) observations by the CEERS and GLASS Early Release Science (ERS) programs. Given that all the galaxies are located in well-studied fields with stars boasting precise astrometry, we have not included target acquisition. Any small inaccuracies in acquisition are inconsequential due to the 3"x3" NIRSpec-IFS Field of View. For maximizing the observations schedulability, we have not set any constraints on the PA, except for one galaxy. Additionally, our dither strategy is designed to minimize issues related to MSA leakage, rendering a leakcal exposure unnecessary.

Since we want to reach the same depth in $\text{SFR} = 1.5 \text{ Msun/yr}$ for the clumps, we set slightly different exposure times depending on redshift to reach a similar S/N on H β and HeII 1640 (> 5 in all cases). We employ a small cycling pattern with 9-point dithers (on average) to improve spatial and spectral sampling, as suggested in the JWST documentation. The small cycling also does not significantly impact the field of view with complete exposure time ($\sim 2.8'' \times 2.8''$). We end with a total exposure time of 49 hours (on source) to observe the entire sample with both gratings.

No background exposures are necessary since each observation comprises numerous spaxels that are uncontaminated by the emission from the central source or neighboring sources. We will make use of spaxels within the IFS Field of View that lack line emission to execute background subtraction.

Proposal 7014 - Targets - Searching for Population III stars around massive interacting galaxies in the reionization epoch

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	ID_1	RA: 00 14 28.1280 (3.6172000d) Dec: -30 25 31.91 (-30.42553d) Equinox: J2000 Comments: DHZ1 Category=Galaxy Description=[High-redshift galaxies]		
	(2)	ID_2	RA: 00 14 24.8600 (3.6035833d) Dec: -30 22 55.75 (-30.38215d) Equinox: J2000 Comments: YD7 + YD4 + YD6 + YD1 Category=Galaxy Description=[High-redshift galaxies] Extended=YES		
	(3)	ID_3	RA: 00 14 25.6095 (3.6067063d) Dec: -30 22 51.05 (-30.38085d) Equinox: J2000 Comments: ZD6 + other clumpy galaxies - pointing centered in the barycenter of the system Category=Galaxy Description=[High-redshift galaxies]		
	(4)	ID_4_5	RA: 14 20 1.2100 (215.0050417d) Dec: +52 59 47.40 (52.99650d) Equinox: J2000 Comments: ID CEERS 3 and 4 - slightly moving toward south - west to include other galaxies/clumps that might be at the same redshift. Category=Galaxy Description=[High-redshift galaxies]		
	(5)	ID_6	RA: 14 19 19.3778 (214.8307408d) Dec: +52 53 15.86 (52.88774d) Equinox: J2000 Comments: ID CEERS 20 - Pointing slightly offset downwards to get farther from a low-z interloper in the upper part. Category=Galaxy Description=[High-redshift galaxies]		
	(6)	ID_7	RA: 14 20 0.6250 (215.0026042d) Dec: +53 00 27.00 (53.00750d) Equinox: J2000 Comments: ID CEERS 749 - slightly shifted to be closer to a possible neighbouring candidate at the same redshift. Category=Galaxy Description=[High-redshift galaxies]		
	(7)	ID_8	RA: 14 19 26.7250 (214.8613542d) Dec: +52 52 33.50 (52.87597d) Equinox: J2000 Comments: CEERS ID 829 -slightly shifted the center to avoid bright galaxy on the left, and to marginally include a merging system at $z = 2.5$. Category=Galaxy Description=[High-redshift galaxies]		
	(8)	ID_9	RA: 14 20 8.4500 (215.0352083d) Dec: +52 53 26.10 (52.89058d) Equinox: J2000 Comments: ID 1019 - slightly shited the center to avoid nearby bright galaxies Category=Galaxy Description=[High-redshift galaxies]		

Proposal 7014 - Observation 1 - Searching for Population III stars around massive interacting galaxies in the reionization epoch

Observation	Proposal 7014, Observation 1											Fri May 15 18:00:11 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	ID_1	RA: 00 14 28.1280 (3.6172000d) Dec: -30 25 31.91 (-30.42553d) Equinox: J2000									
	Comments: DHZ1 Category=Galaxy Description=[High-redshift galaxies]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			MEDIUM		1		5			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140M/F100LP	NRSIRS2	22	2	false	true	NONE	5	10	16193.668	

Proposal 7014 - Observation 2 - Searching for Population III stars around massive interacting galaxies in the reionization epoch

Observation	Proposal 7014, Observation 2											Fri May 15 18:00:11 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(2)	ID_2	RA: 00 14 24.8600 (3.6035833d) Dec: -30 22 55.75 (-30.38215d) Equinox: J2000									
	Comments: YD7 + YD4 + YD6 + YD1 Category=Galaxy Description=[High-redshift galaxies] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		MEDIUM		1		5				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140M/F100LP	NRSIRS2	17	2	false	true	NONE	5	10	12546.445	
	2	G395M/F290LP	NRSIRS2	23	2	false	true	NONE	5	10	16923.112	

Proposal 7014 - Observation 3 - Searching for Population III stars around massive interacting galaxies in the reionization epoch

Observation	Proposal 7014, Observation 3												Fri May 15 18:00:11 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(3)	ID_3		RA: 00 14 25.6095 (3.6067063d)									
				Dec: -30 22 51.05 (-30.38085d)									
				Equinox: J2000									
		Comments: ZD6 + other clumpy galaxies - pointing centered in the barycenter of the system Category=Galaxy Description=[High-redshift galaxies]											
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1	CYCLING			MEDIUM		1		5				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G140M/F100LP	NRSIRS2	17	2	false	true	NONE	5	10	12546.445		

Proposal 7014 - Observation 4 - Searching for Population III stars around massive interacting galaxies in the reionization epoch

Observation	Proposal 7014, Observation 4												Fri May 15 18:00:11 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: NIRSspec IFU Spectroscopy												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(4)	ID_4_5		RA: 14 20 1.2100 (215.0050417d)									
				Dec: +52 59 47.40 (52.99650d)									
				Equinox: J2000									
		Comments: ID CEERS 3 and 4 - slightly moving toward south - west to include other galaxies/clumps that might be at the same redshift. Category=Galaxy Description=[High-redshift galaxies]											
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1	CYCLING			MEDIUM		1		5				
Spectral Elements	#	Grating/Filter		Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140M/F100LP		NRSIRS2	17	2	false	true	NONE	5	10	12546.445	
	2	G395M/F290LP		NRSIRS2	23	2	false	true	NONE	5	10	16923.112	
Special Requirements	Aperture PA Range 82.97164917 to 134.97164917 Degrees (V3 304.0 to 356.0) Aperture PA Range 173.97164917 to 45.97164917 Degrees (V3 35.0 to 267.0)												

Proposal 7014 - Observation 5 - Searching for Population III stars around massive interacting galaxies in the reionization epoch

Observation	Proposal 7014, Observation 5												Fri May 15 18:00:11 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(5)	ID_6	RA: 14 19 19.3778 (214.8307408d) Dec: +52 53 15.86 (52.88774d) Equinox: J2000 Comments: ID CEERS 20 - Pointing slightly offset downwards to get farther from a low-z interloper in the upper part. Category=Galaxy Description=[High-redshift galaxies]										
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	CYCLING		MEDIUM		1		5					
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G140M/F100LP	NRSIRS2	17	2	false	true	NONE	5	10	12546.445		
	2	G395M/F290LP	NRSIRS2	23	2	false	true	NONE	5	10	16923.112		

Proposal 7014 - Observation 6 - Searching for Population III stars around massive interacting galaxies in the reionization epoch

Observation	Proposal 7014, Observation 6												Fri May 15 18:00:11 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(6)	ID_7	RA: 14 20 0.6250 (215.0026042d) Dec: +53 00 27.00 (53.00750d) Equinox: J2000 <i>Comments: ID CEERS 749 - slightly shifted to be closer to a possible neighbouring candidate at the same redshift.</i> <i>Category=Galaxy</i> <i>Description=[High-redshift galaxies]</i>										
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1	CYCLING			MEDIUM		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G140M/F100LP	NRSIRS2	13	1	false	true	NONE	9	9	8665.801		
	2	G395M/F290LP	NRSIRS2	20	1	false	true	NONE	9	9	13261.301		

Proposal 7014 - Observation 7 - Searching for Population III stars around massive interacting galaxies in the reionization epoch

Observation	Proposal 7014, Observation 7												Fri May 15 18:00:11 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(7)	ID_8		RA: 14 19 26.7250 (214.8613542d)									
				Dec: +52 52 33.50 (52.87597d)									
				Equinox: J2000									
	Comments: CEERS ID 829 -slightly shifted the center to avoid bright galaxy on the left, and to marginally include a merging system at z = 2.5.												
	Category=Galaxy												
	Description=[High-redshift galaxies]												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1	CYCLING			MEDIUM		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G140M/F100LP	NRSIRS2	14	1	false	true	NONE	9	9	9322.301		
	2	G395M/F290LP	NRSIRS2	21	1	false	true	NONE	9	9	13917.801		

Proposal 7014 - Observation 8 - Searching for Population III stars around massive interacting galaxies in the reionization epoch

Observation	Proposal 7014, Observation 8												Fri May 15 18:00:11 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: NIRSpect IFU Spectroscopy												
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(8)	ID_9		RA: 14 20 8.4500 (215.0352083d)									
				Dec: +52 53 26.10 (52.89058d)									
				Equinox: J2000									
		Comments: ID 1019 - slightly shited the center to avoid nearby bright galaxies Category=Galaxy Description=[High-redshift galaxies]											
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1	CYCLING			MEDIUM		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G140M/F100LP	NRSIRS2	22	1	false	true	NONE	9	9	14574.301		